Script in AI

- **Definition**: A script is a predefined framework that represents a typical sequence of actions in a familiar context (e.g., dining at a restaurant, visiting a doctor).
- **Origin**: Developed from cognitive psychology to explain how humans recall and anticipate events.
- **Purpose in AI**: Enables systems to interpret incomplete information by filling gaps with expected events.

Structure of a Script

Scripts usually contain:

- **Entry conditions**: What must be true before the script starts (e.g., being hungry before dining).
- **Roles**: Participants involved (e.g., customer, waiter, cashier).
- **Props**: Objects used (e.g., menu, food, bill).
- **Scenes**: Ordered steps (e.g., entering restaurant → reading menu → ordering → eating → paying).
- **Results**: Expected outcomes (e.g., customer is no longer hungry, restaurant earns money).

Why Scripts Matter in AI

- **Natural Language Understanding**: Helps AI infer meaning when information is incomplete. For example, if someone says "I went to a restaurant," the AI can infer that ordering, eating, and paying likely occurred.
- **Common-Sense Reasoning**: Provides background knowledge for everyday scenarios.
- **Efficiency**: Reduces complexity by using stereotypical event sequences instead of modeling every possible variation.
- **Applications**:
    - Chatbots and virtual assistants
    - Story comprehension systems
    - Autonomous agents in simulations

- Knowledge-based expert systems

Comparison: Scripts vs. Other Knowledge Representations

Representation	Focus	Strengths	Limitations
Scripts	Sequences of stereotypical events	Good for everyday scenarios, fills gaps in incomplete info	Rigid; struggles with novel or unusual events
Frames	Objects and their attributes	Flexible, hierarchical	Less suited for temporal sequences
Semantic Networks	Relationships between concepts	Easy visualization of connections	Weak in handling event order
Production Rules	IF-THEN logic	Strong for reasoning and decision-making	Can become complex with large rule

Challenges & Limitations

- **Rigidity**: Scripts assume stereotypical behavior, so they may fail in unusual or unexpected situations.
- **Cultural Bias**: Scripts are context-dependent; a "restaurant script" in India may differ from one in the U.S.
- **Scalability**: Creating and maintaining a large library of scripts is resource-intensive.

Examples

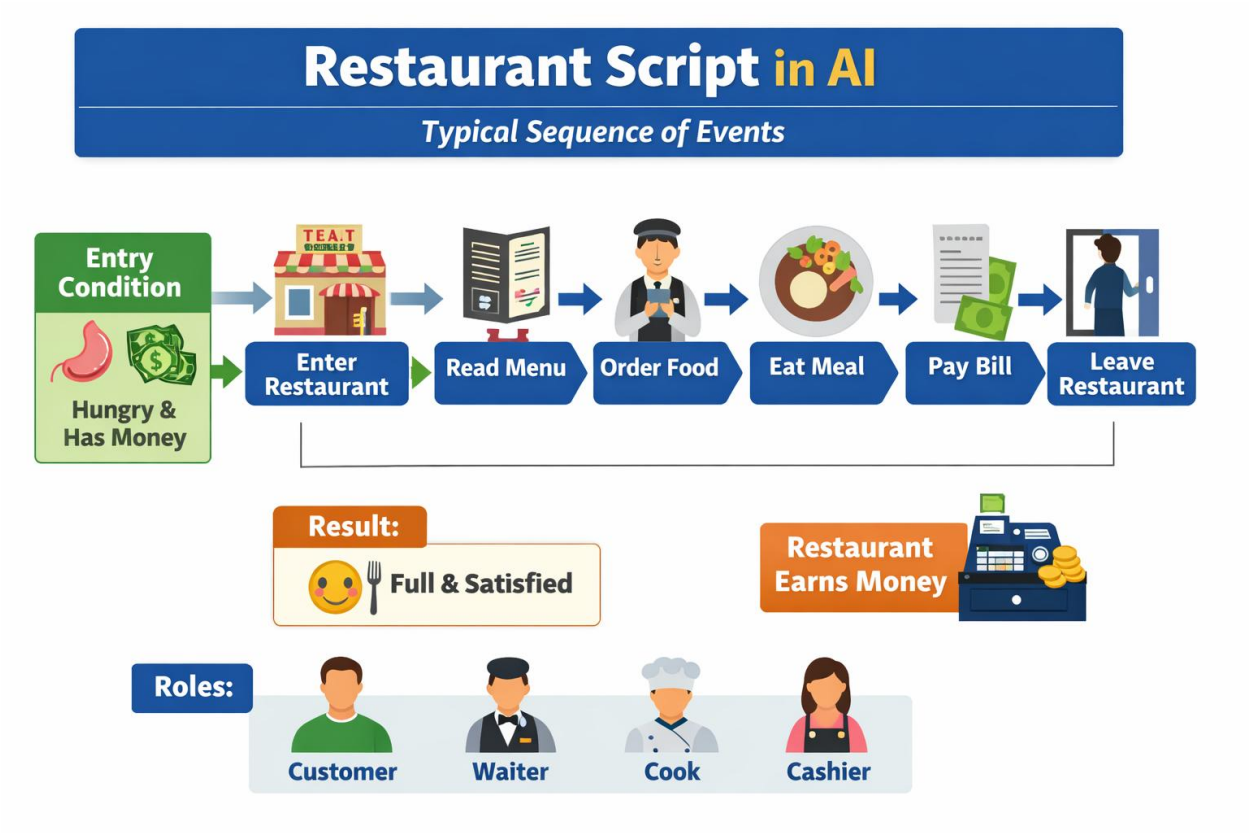
Restaurant Script

- Entry condition: Customer is hungry and has money.
- Roles: Customer, waiter, cook, cashier.
- Props: Menu, table, food, bill, money.
- Scenes:
    1. Customer enters restaurant.

2. Customer is seated and receives menu.
3. Customer orders food from waiter.
4. Cook prepares food.
5. Waiter serves food.
6. Customer eats.
7. Customer pays bill.
8. Customer leaves.

- Result: Customer is no longer hungry, restaurant earns money.

For instance, if someone says “I went to a restaurant,” the AI can infer that ordering, eating, and paying likely happened, even if not explicitly mentioned.



Movie Theater Script

- Entry condition: Person wants entertainment and has money.
- Roles: Customer, ticket seller, usher.

- Props: Ticket, popcorn, seat.
- Scenes:
    1. Customer buys ticket.
    2. Customer enters theater.
    3. Customer finds seat.
    4. Movie plays.
    5. Customer watches movie.
    6. Customer leaves theater.
- Result: Customer is entertained.

Doctor Visit Script

- Entry condition: Patient feels unwell.
- Roles: Patient, doctor, nurse, receptionist.
- Props: Appointment slip, medical instruments, prescription.
- Scenes:
    1. Patient arrives at clinic.
    2. Patient checks in with receptionist.
    3. Patient waits in lobby.
    4. Doctor examines patient.
    5. Doctor prescribes medicine or treatment.
    6. Patient leaves clinic.
- Result: Patient receives medical advice or treatment.

Airport Script

- Entry condition: Traveler wants to fly to another city.
- Roles: Passenger, airline staff, security officer.
- Props: Ticket, luggage, boarding pass.

- Scenes:
    1. Passenger arrives at airport.
    2. Passenger checks in luggage.
    3. Passenger passes security check.
    4. Passenger waits at gate.
    5. Passenger boards plane.
    6. Plane takes off and lands.
    7. Passenger collects luggage.
- Result: Passenger reaches destination.